

Supplementary Information: Cycle-Consistent Safety-Oriented Dual-LSTM for Aero-Engine RUL Prediction

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1 Experiment Matrix

The full matrix contains four C-MAPSS subsets, three seeds, and four jobs: LSTM baseline, Dual-LSTM no-cycle, Dual-LSTM cycle, and Dual-LSTM cycle+safety-w2. Each job writes metrics, predictions, training history, and selected-feature files. The paired-window horizon is $k = 1$, the window size is 30, and the RUL cap is 130.

2 Full Three-Seed Test Summary

Table 1: Full Paper 2 three-seed test summary. Lower values are better.

Subset	Job	RMSE	MAE	Critical RMSE50	OverRatio	OverMag
FD001	LSTM baseline	14.98	11.34	5.83	0.630	7.17
FD001	Dual-LSTM no-cycle	15.51	11.61	6.86	0.623	6.97
FD001	Dual-LSTM cycle	14.86	11.13	5.96	0.653	7.37
FD001	Dual-LSTM cycle safety-w2	14.88	10.66	5.39	0.487	4.23
FD002	LSTM baseline	18.38	13.58	8.87	0.387	4.61
FD002	Dual-LSTM no-cycle	17.95	13.35	9.06	0.399	5.12
FD002	Dual-LSTM cycle	18.06	13.52	9.54	0.430	5.50
FD002	Dual-LSTM cycle safety-w2	20.32	15.35	8.82	0.347	3.83
FD003	LSTM baseline	14.71	10.49	5.98	0.567	6.76
FD003	Dual-LSTM no-cycle	15.34	11.04	6.08	0.607	7.55
FD003	Dual-LSTM cycle	14.77	10.51	5.48	0.557	6.56
FD003	Dual-LSTM cycle safety-w2	14.37	10.00	4.33	0.427	4.65
FD004	LSTM baseline	23.40	17.50	22.01	0.512	9.20
FD004	Dual-LSTM no-cycle	22.43	16.80	19.97	0.538	9.11
FD004	Dual-LSTM cycle	22.32	16.54	19.19	0.548	9.00
FD004	Dual-LSTM cycle safety-w2	22.38	16.57	15.11	0.445	5.96

3 Paired Seed-Engine Bootstrap

The bootstrap pairs predictions by subset, seed, and test engine. The table reports candidate minus baseline. Negative values indicate lower error for the candidate job.

Table 2: Bootstrap comparison table for the main Paper 2 contrasts.

Subset	Contrast	Metric	Difference	95% CI	Interpretation
FD001	cycle_vs_lstm	rmse	-0.13	[-0.91, 0.65]	uncertain
FD001	cycle_vs_lstm	critical_rmse_50	0.13	[-0.87, 1.31]	uncertain
FD001	cycle_vs_lstm	overestimation_magnitude	0.20	[-0.33, 0.75]	uncertain
FD001	safety_vs_cycle	rmse	0.02	[-0.94, 1.06]	uncertain
FD001	safety_vs_cycle	critical_rmse_50	-0.49	[-1.35, 0.29]	uncertain
FD001	safety_vs_cycle	overestimation_magnitude	-3.14	[-3.78, -2.51]	improvement
FD002	cycle_vs_lstm	rmse	-0.33	[-1.05, 0.33]	uncertain
FD002	cycle_vs_lstm	critical_rmse_50	0.67	[-0.50, 1.98]	uncertain
FD002	cycle_vs_lstm	overestimation_magnitude	0.90	[0.56, 1.23]	cost_or_worse
FD002	safety_vs_cycle	rmse	2.28	[1.52, 3.03]	cost_or_worse
FD002	safety_vs_cycle	critical_rmse_50	-0.68	[-1.94, 0.54]	uncertain
FD002	safety_vs_cycle	overestimation_magnitude	-1.67	[-2.04, -1.33]	improvement
FD003	cycle_vs_lstm	rmse	0.06	[-1.09, 1.17]	uncertain
FD003	cycle_vs_lstm	critical_rmse_50	-0.52	[-2.19, 0.89]	uncertain
FD003	cycle_vs_lstm	overestimation_magnitude	-0.20	[-0.86, 0.45]	uncertain
FD003	safety_vs_cycle	rmse	-0.41	[-1.65, 0.75]	uncertain
FD003	safety_vs_cycle	critical_rmse_50	-1.20	[-2.17, -0.30]	improvement
FD003	safety_vs_cycle	overestimation_magnitude	-1.91	[-2.57, -1.28]	improvement
FD004	cycle_vs_lstm	rmse	-1.08	[-1.97, -0.31]	improvement

Subset	Contrast	Metric	Difference	95% CI	Interpretation
FD004	cycle_vs_lstm	critical_rmse_50	-2.84	[-4.28, -1.48]	improvement
FD004	cycle_vs_lstm	overestimation_magnitude	-0.20	[-0.67, 0.21]	uncertain
FD004	safety_vs_cycle	rmse	0.07	[-1.02, 1.15]	uncertain
FD004	safety_vs_cycle	critical_rmse_50	-4.46	[-6.29, -2.50]	improvement
FD004	safety_vs_cycle	overestimation_magnitude	-3.04	[-3.63, -2.49]	improvement

4 Paper 1 to Paper 2 Bridge

This bridge compares the best Paper 2 job with the best Paper 1 representative-matrix value. It is designed to prevent overclaiming.

Table 3: Paper 1 representative matrix versus Paper 2 best job. Negative change means Paper 2 is lower.

Subset	Metric	Paper1 best	Paper1 value	Paper2 best	Paper2 value	Change
FD001	rmse	Gradient Boosting	12.81	Dual-LSTM cycle	14.86	+15.9%
FD001	critical_rmse_50	Safety-GRU	4.86	Dual-LSTM cycle safety-w2	5.39	+10.9%
FD001	overestimation_magnitude	Safety-GRU	4.79	Dual-LSTM cycle safety-w2	4.23	-11.8%
FD002	rmse	GRU	17.81	Dual-LSTM no-cycle	17.95	+0.8%
FD002	critical_rmse_50	Safety-GRU	8.12	Dual-LSTM cycle safety-w2	8.82	+8.6%
FD002	overestimation_magnitude	Safety-GRU	3.41	Dual-LSTM cycle safety-w2	3.83	+12.4%
FD003	rmse	Gradient Boosting	13.21	Dual-LSTM cycle safety-w2	14.37	+8.8%
FD003	critical_rmse_50	Random Forest	4.76	Dual-LSTM cycle safety-w2	4.33	-9.0%
FD003	overestimation_magnitude	Safety-GRU	4.84	Dual-LSTM cycle safety-w2	4.65	-3.9%
FD004	rmse	Random Forest	20.18	Dual-LSTM cycle	22.32	+10.6%
FD004	critical_rmse_50	Safety-GRU	14.62	Dual-LSTM cycle safety-w2	15.11	+3.3%
FD004	overestimation_magnitude	Safety-GRU	5.51	Dual-LSTM cycle safety-w2	5.96	+8.2%

5 Supplementary Figure Index

All figures are generated by Python/matplotlib from CSV outputs. Editable SVG and PDF files are retained, with PNG/TIFF previews.

- Figure 1: Dual-LSTM architecture and inference boundary.
- Figure 2: metric-rank heatmap.
- Figure 3: RMSE versus Critical RMSE50.
- Figure 4: RMSE versus overestimation magnitude.
- Figure 5: Paper 1 to Paper 2 bridge.

6 Claim Boundaries

This supplementary file follows the same claim boundaries as the main manuscript: no first-ever Dual-LSTM RUL claim, no full-portfolio SOTA claim, no aviation safety certification claim, and no real-fleet validation claim.